Consensus modelling

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LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Background

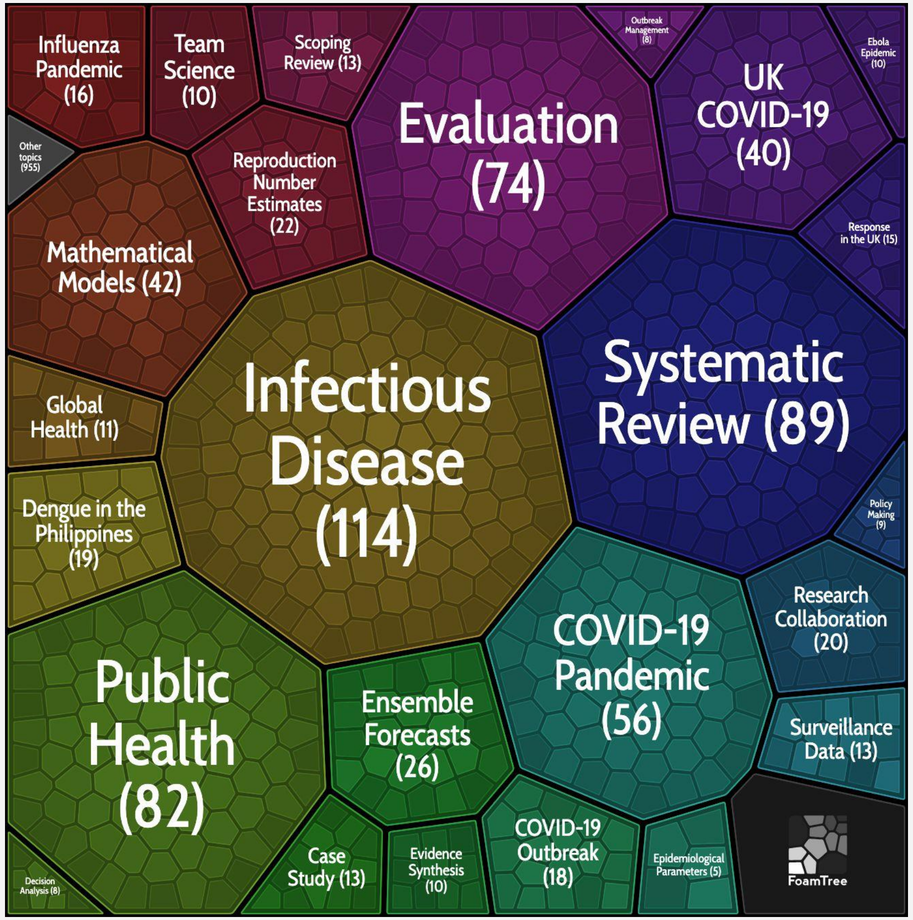
Crisis, collaboration, comparison

- Hazards, epi, dengue, COVID-19
- Rt estimation, forecasting, ensembles
- Multi-model infrastructure (UK; US; EU)

Model comparison <> Evaluation <> Ethics

Research interest

- Evaluating real-time modelling
  - Quality, comparison, combination
  - Value, decision relevance
  - Trustworthiness, responsibility



Titles in my zotero library, clustered with carrot2

Modelling in real-time emergencies

Many **uncertainties** of an evolving outbreak:

- **Mechanisms**: unobserved, heterogeneous
- **Data**: lagged, sparse
- **Decisions**: timeliness, stakes

Modelling offers **synthesis**:

...expert judgement

...available data

...for a target context

UK method:
consensus modelling via SPI-M-O

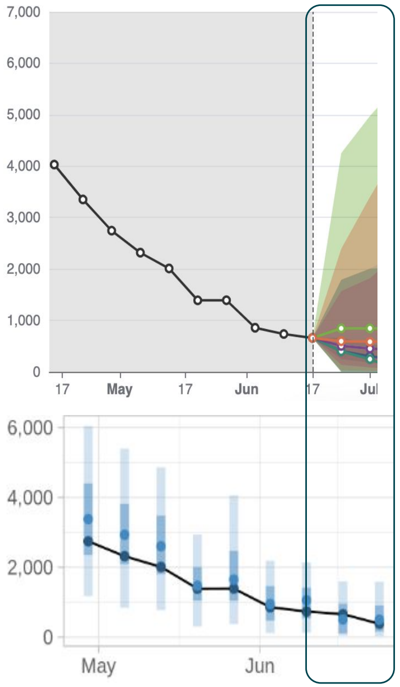
Conclusions

Consensus is important when producing evidence to support decisions in any situation, but during a pandemic becomes critical. The process of creating a consensus provides multiple functions: peer-review of methods and approach, emphasis of uncertainty, shared responsibility as protection to individuals and development of a shared purpose and the consequent emotional support. Whilst the process sometimes felt too slow and cumbersome, it is worth the outcome.

Medley 2022 https://doi.org/10.1016/j.jbior.2022.100918

Model comparison: Short term forecasts

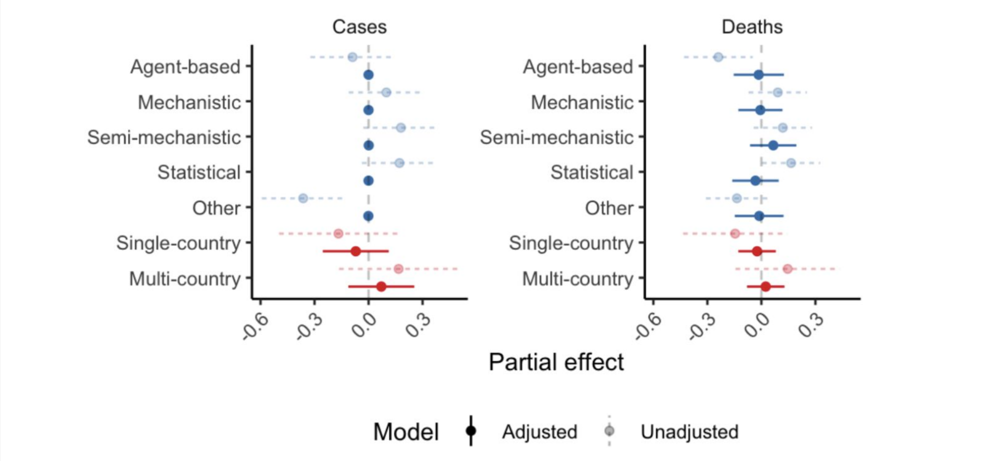
Ensembles perform reliably well…



COVID-19 hospitalisations in Germany, May-July 2023: forecasts from all models, and accuracy of the Hub ensemble (below)

Sherratt, K., and others, 2023, eLife, doi.org/10.7554/eLife.81916

… and model structure doesn't appear to matter much for **overall predictive accuracy**



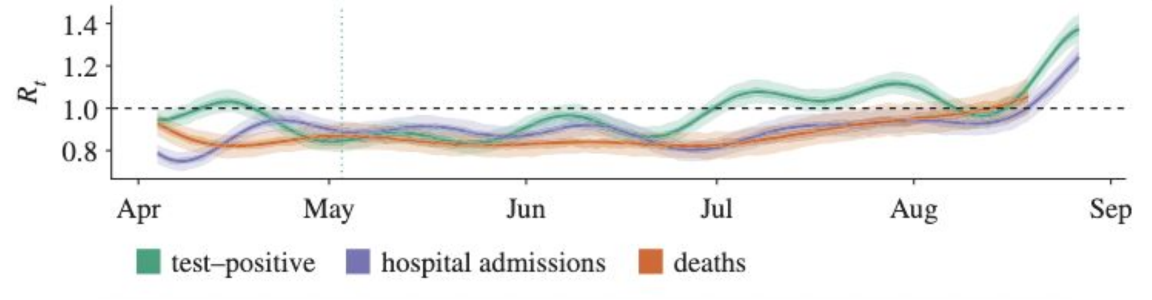
47 models, 32 countries, 181,851 probabilistic predictions: modelling overall accuracy (interval score) by model structure

Sherratt, K and others, 2025, medrXiv, doi.org/10.1101/2025.04.10.25325611

Model comparison: Reproduction numbers

- The reproduction number (Rt) represents underlying transmission process
- This makes modelling **very sensitive to assumptions** - and estimates difficult to evaluate objectively
- Even a *single* consistent model might **blur different signals** from heterogeneous transmission

Then comparing across models with different data, delay estimates, reference dates...

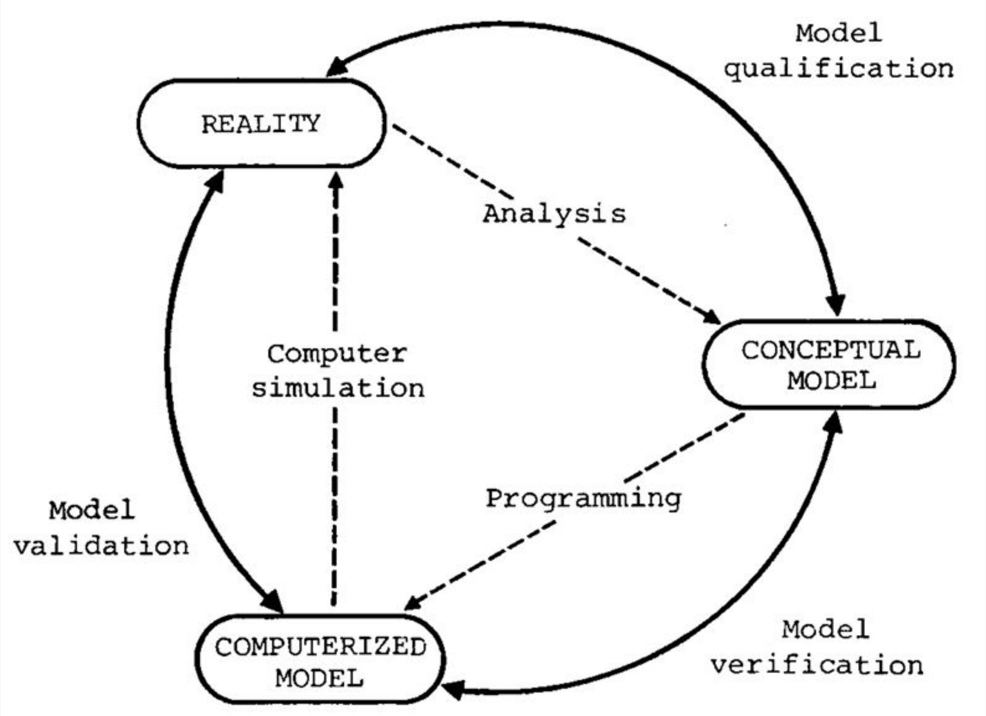


Estimates of Rt in England: one model fitting to three epidemic indicators, April-September 2020. Showing daily Rt estimate (90% CI), from: all test-positive cases, hospital admissions, and deaths with a positive test in the previous 28 days. Sherratt and others 2021, Phil. Trans. R. Soc. B, https://doi.org/10.1098/rstb.2020.0283

Model evaluation

Evaluating one model…

- Qualification | *Assumptions*
- Verification | *Implementation*
- Validation | *Calibration*



Schlesinger 1979

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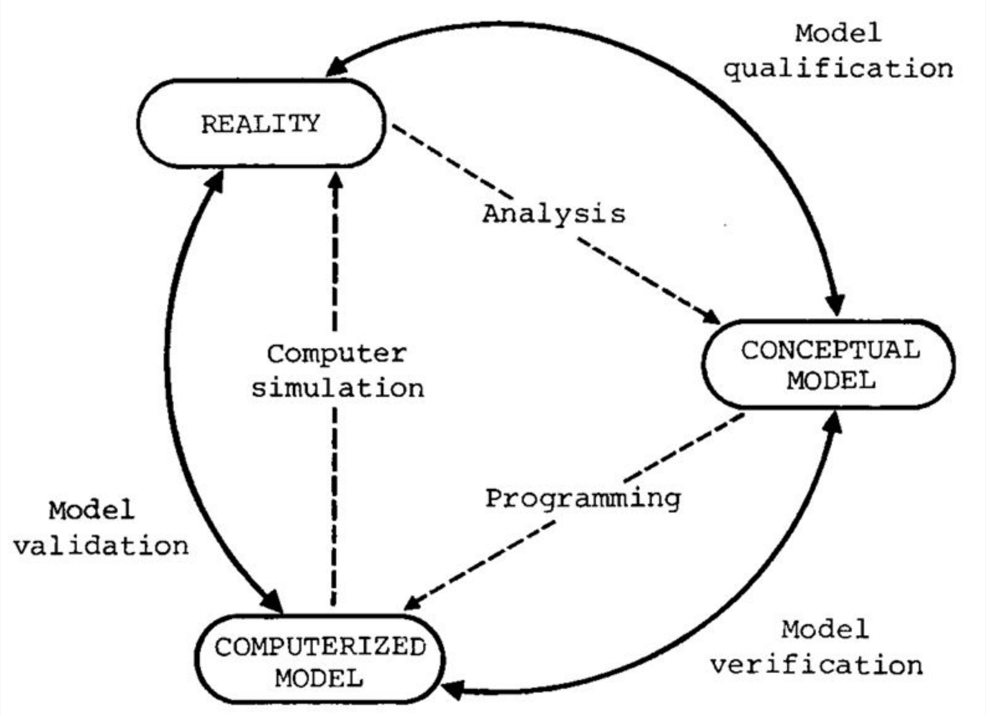


Schlesinger 1979

Can we apply this to SPI-M "consensus model"?

Model evaluation: SPI-M-O consensus

- Qualification

... context and structure

evidence/synthesis or advice/strategy?

- Verification

... group consensus-making processes

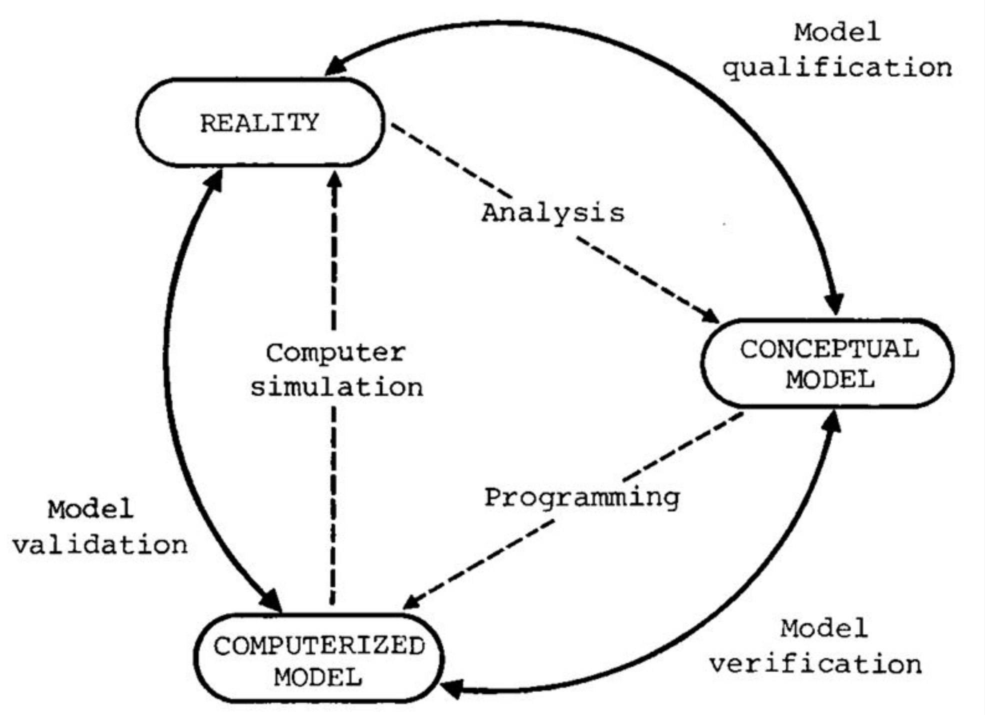
expert elicitation?

- Validation

... consensus statement outputs

R, growth rate, projections?

wrong **tell me why**



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Schlesinger 1979

Evaluating modelling

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Evaluation > model credibility

- **Trust**: professionalism, responsibility

What is the **professional** network of academic ID modelling?

Where are the gaps?

- **Value**: decision relevance, engagement

Who are the decision-makers in **routine** outbreak response?

What are the decision criteria?

Framework: TQV and the Code Principles

Trustworthiness

- About building confidence in the integrity and professionalism of producers
- Requires transparency, ethical behaviour, and responsible leadership

Quality

- Use appropriate data and methods to produce reliable statistics
- Focus on continuous improvement and be open about limitations

Value

- Ensure statistics are relevant, accessible, and support decision-making
- Engage with users and be responsive to their needs

Code Principles (10 in total)

Trustworthiness	Quality	Value
1. Show integrity	5. Prioritise quality	8. Be relevant
2. Lead responsibly	6. Be rigorous	9. Be clear
3. Be transparent	7. Be open about quality	10. Be accessible
4. Manage data responsibly		

Office for Statistics Regulation

Use the QR code to visit the Code of Practice website to find out more

Many thanks

Conclusions

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Lived experience